

Customer Survival -- SaaS Subscription Chu

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2026-02-26

Mode: Full Pipeline

1. Problem Formulation

Problem Type: Find
Domain: Statistical Inference

Universe

- Kaplan Meier: $\{\text{median_basic}, \text{median_pro}\}$
- Logrank Test: $\{\text{statistic}, \text{p_value}\}$
- Ph Assumption: $\{\text{plan_schoenfeld_p}, \text{tickets_schoenfeld_p}, \text{holds}\}$

Variables

Symbol	Meaning	Type / Range
μ_1, μ_2	population means for groups 1 and 2	$\mathbb{R}$
$\bar{x}_1, \bar{x}_2$	sample means	$\mathbb{R}$
s_1, s_2	sample standard deviations	$\mathbb{R}_{>0}$
n_1, n_2	sample sizes	$\mathbb{Z}^+$
t	test statistic (hypothesis test)	$\mathbb{R}$
p	p-value	$[0, 1]$
α	significance level	$[0, 1]$

Structure

- **Customers**: 500 SaaS subscribers with subscription duration (months) and churn status
- **Censoring**: ~30% of customers are right-censored (still active, have not churned)
- **Plan types**: Basic (n=300, median survival ~12 months) and Pro (n=200, median survival ~24 months)
- **Covariate**: Support ticket count (higher tickets correlate with higher churn risk)
- **Questions**: Do Basic and Pro plans have different churn rates? What factors predict churn?

Real-World Mapping

Real-World Concept	Mathematical Object
group average	sample mean $\bar{x}$

group variability
sample count
significance threshold

sample std s
n
alpha = 0.05

Constraints

(1)

$$H_0 : \mu_1 = \mu_2$$

null hypothesis

(2)

$$H_1 : \mu_1 \neq \mu_2$$

alternative hypothesis

(3)

$$\alpha = 0.05$$

significance level

(4)

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Welch's t-test statistic

Objective

Reject H_0 if $p < \alpha$

Approach

Suggested approach Kaplan-Meier + Log-Rank Test + Cox Proportional Hazards

Complexity class:

Available tools: Kaplan-Meier + Log-Rank Test + Cox Proportional Hazards

2. Solution

Answer:	Statistical test complete
Optimal:	No / Unknown
Feasible:	Yes
Algorithm:	Kaplan-Meier + Log-Rank Test + Cox Proportional Hazards
Time:	0.0964s
Certificate:	Assumptions checked; test statistic independently verified.

Solution Details

Statistical analysis computed.

Verification

✓	Km Basic Monotonic	Passed
✓	Km Pro Monotonic	Passed
✓	Km Basic In 01	Passed
✓	Km Pro In 01	Passed
✓	Logrank Significant	Passed
✓	Cox Plan Hr Gt 1	Passed
✓	Cox Tickets Hr Gt 1	Passed
✓	Concordance Gt 055	Passed
✓	Median Basic Lt Pro	Passed
✓	Censoring Rate Approx 30Pct	Passed
✓	All Passed	Passed

3. Interpretation

The Question

- **Customers**: 500 SaaS subscribers with subscription duration (months) and churn status
- **Censoring**: ~30% of customers are right-censored (still active, have not churned)
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The Answer

Test complete.

What This Means

- Kaplan-Meier: Monotonically decreasing survival curves for Basic and Pro plans
 - Median survival: Basic ~12 months, Pro ~24 months
 - Log-rank test: $p < 0.05$ (significant difference between plans)
 - Cox PH hazard ratios: Basic vs Pro HR > 1.0 (Basic has higher hazard), tickets HR > 1.0 (more tickets = higher churn)
 - 95% CI: Hazard ratio confidence intervals excluding 1.0 for both covariates
 - Concordance index: > 0.55 (better than random prediction)
 - PH assumption: Schoenfeld residuals test for proportional hazards
1. Kaplan-Meier estimator: Non-parametric survival function estimate per plan group, handling right-censored observations
 2. Log-rank test: Non-parametric hypothesis test comparing survival distributions between Basic and Pro plans
 3. Cox Proportional Hazards: Semi-parametric regression modeling hazard as a function of plan type and support ticket count; reports hazard ratios with 95% confidence intervals
 4. Diagnostics: Proportional hazards assumption check via Schoenfeld residuals; concordance index for predictive accuracy

Recommendations

1. Report effect sizes alongside p-values for practical significance assessment.

Limitations

- Results are conditional on model assumptions (normality, independence, etc.).

- Statist